$$\frac{\partial}{\partial t}\rho_{\mathcal{S}}(X,X',t) = \Bigg[-\frac{\mathrm{i}}{2M}\left(\frac{\partial^2}{\partial X'^2}-\frac{\partial^2}{\partial X^2}\right)-\frac{\mathrm{i}}{2}M\left(\Omega^2+\widetilde{\Omega}^2\right)\left(X^2-X'^2\right)$$
$$+\gamma(X-X')\left(\frac{\partial}{\partial X'}-\frac{\partial}{\partial X}\right)-D(X-X')^2$$
$$+\mathrm{i}f(X-X')\left(\frac{\partial}{\partial X'}+\frac{\partial}{\partial X}\right)\Bigg]\rho_{\mathcal{S}}(X,X',t). \qquad (5.28)$$